Uncertainty Signal Pilot: Perception and Reasoning Entropy

FERMAT Handwritten Math, Qwen2.5-VL-3B

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Abstract

This pilot tests whether two entropy-based uncertainty signals – *perception entropy* (instability in transcribing a handwritten math answer) and *reasoning entropy* (instability in grading whether an answer contains an error) – carry real signal about model correctness, on 100 items from the FERMAT dataset with Qwen2.5-VL-3B. **Perception entropy is the one result that resolves statistically**: AUROC = 0.788, 95% CI $[0.697, 0.869]$, and it survives every artifact check applied to it – removing all parse failures, all max-entropy items, and both together each leave the CI excluding chance (0.707, $[0.587, 0.819]$, at $n = 74$ under the harshest cut). Reasoning entropy does not resolve at this sample size. Its digit-only signal is AUROC = 0.618 at $K = 5$ with a CI of $[0.489, 0.741]$ – *including* chance. Resampling to $K = 15$ raised the point estimate to 0.665 $[0.514, 0.800]$, but a paired test shows that change is not distinguishable from zero. A follow-up experiment clustering the model's own `**Reasoning:**` explanation text rather than its parsed `**Error:**` digit reached 0.702 $[0.579, 0.817]$ at $K = 5$; that value does exclude chance on its own, but the *paired* difference against the digit-only baseline it was meant to beat is $+0.083$ $[-0.064, +0.233]$ – so the improvement itself is unproven. It also has a real mechanical defect: at $K = 15$ it falls to 0.520, and the paired $K = 5$ vs. $K = 15$ degradation ($+0.182$ in favour of $K = 5$, $[+0.037, +0.330]$) is the one reasoning-arm comparison that does survive, consistent with a diagnosed transitive cascade over-merging bug in the clustering algorithm. The honest summary is that perception entropy shows a clear association with errors, reasoning entropy is directionally suggestive but not statistically resolved at $n = 100$, and $K = 15$ provides no reliable improvement at this sample size. Sample size, not technique, is the binding constraint on every open question here.

1 Goal

Given a vision-language model transcribing and grading handwritten math answers, does the entropy of its own repeated outputs – how much it disagrees with itself across independent samples – predict when it is actually wrong? This is a fast, scoped-down pilot to check whether either signal exists at all, not the full study: no baseline suite, no conformal calibration, no human double grading, no full-dataset run.

Every AUROC below is reported with a 95% bootstrap confidence interval (10,000 item-level resamples, seed 0; `pilot.plotting.bootstrap_auroc_ci`). These were added late, after the substantive experiments were complete, and they materially changed what this report can claim – several differences it originally presented as findings turn out to be within sampling error at $n = 100$. Where two conditions are compared, the comparison uses a *paired* bootstrap over the same resampled items (`bootstrap_auroc_difference_ci`) rather than checking whether two marginal intervals overlap, which is a strictly weaker and commonly misread test.

2 Setup

- **Data**: 100 items sampled (seed 42) from the FERMAT dataset.
- **Model**: Qwen2.5-VL-3B-Instruct.
- **Perception**: the model transcribes a handwritten Question–Answer pair; $K = 5$ samples at temperature 0.7, plus 2 samples at temperature 0 as a determinism sanity anchor.
- **Reasoning**: the model grades whether the handwritten Answer contains an error (binary 0/1); same $K = 5 + 2$ temperature-0 sampling scheme.
- **Entropy**: Shannon entropy in nats over the cluster distribution of the K sampled outputs for one item, $H = -\sum_i p_i \ln p_i$.
- **Correctness labels**: transcription is compared against the ground-truth answer; grading is compared against the ground-truth error label.

3 Perception Entropy: Two Failures, One Fix

3.1 Attempt 1: exact-string clustering (failed)

The transcription prompt asks for the *full* handwritten derivation, not just a final answer. Clustering by exact string match after light normalization treated any paraphrasing of that derivation as a different answer.

Items pinned at maximum entropy ($\ln 5 \approx 1.609$)	83/100

83% of items showed the maximum possible disagreement regardless of whether the model actually agreed with itself on the substance of the answer – the metric carried essentially no information.

3.2 Attempt 2: naive semantic clustering (failed)

Reasoning that exact-match was too strict, the next attempt clustered by semantic similarity instead: bidirectional NLI entailment and, separately, embedding cosine similarity, both applied to the whole derivation.

	Canonical (Attempt 1)	NLI	Embedding
Mean entropy	1.535	0.137	0.185
Items at max entropy	83/100	0/100	0/100

This looked like a fix – entropy dropped sharply – but the correctness rate that came with it (majority-vote transcription matched against ground truth via the same NLI entailment check) was 92/97 = 95%, implausibly high next to the model's other performance figures. Inspection showed the cause: with premise and hypothesis sharing nearly identical scaffolding (the same LaTeX structure, the same boilerplate phrasing), the NLI model's well-documented lexical-overlap bias pushed it toward "entailment" regardless of whether a buried number or coefficient actually differed. The fix for Attempt 1's problem introduced a new, opposite failure mode: over-merging.

3.3 Attempt 3: final-answer extraction (worked)

The actual defect in both prior attempts was comparing the *wrong span of text*: the full derivation, when only the final answer matters for a correctness judgment. Attempt 3 extracts just the final answer – the last multiple-choice option statement, a `\boxed{}` result, the last display-math block, or (rarest) the last line – before clustering with the same syntactic/canonicalization approach as Attempt 1.

Metric	Value
AUROC (perception entropy $\to$ transcription error)	**0.788**
95% CI	$[0.697, 0.869]$
Items at max entropy	14/100
Transcription correct	42/100
Median entropy, correct group ($n = 42$)	0.500
Median entropy, incorrect group ($n = 58$)	1.332
Mean entropy, correct group	0.583
Mean entropy, incorrect group	1.099

This is the strongest result in the pilot: real, visible separation between correct and incorrect items (Figure 1, left panel), and the only arm whose confidence interval excludes chance by a clear margin.

3.4 Sensitivity: is 0.788 an artifact?

An entropy AUROC can be manufactured two ways that have nothing to do with the model being usefully uncertain, and both are live concerns here. First, parse failures receive their own cluster label, so an unparseable sample inflates entropy *and* tends to be scored incorrect – the metric would then be measuring the parser rather than the model. Second, $K = 5$ entropy takes only 7 distinct values, and the 14 items pinned at $\ln 5$ are *all* incorrect; if that single cell carried the ranking, there would be no graded signal underneath it. Each cut was rerun with its own CI (`pilot.plotting.auroc_sensitivity`):

Subset	n	AUROC	95% CI
Full	100	0.788	$[0.697, 0.869]$
Excluding all parse failures	85	0.773	$[0.672, 0.865]$
Excluding max-entropy items	86	0.721	$[0.611, 0.822]$
Excluding both	74	**0.707**	$[0.587, 0.819]$

Neither cut explains the result. Dropping all 15 parse-failure items moves the estimate by 0.015, and only 3 of the 14 max-entropy items had a parse failure at all (mean entropy 0.99 for parse-failure items vs. 0.864 for clean ones – a small gap, not a confound). Under the harshest simultaneous cut the interval still excludes 0.5. The underlying distribution is monotone rather than concentrated in one cell: of items at zero entropy 11/14 were transcribed correctly, falling steadily to 0/14 at maximum entropy.

That top cell is worth stating as a result rather than treated only as a robustness threat: **every one of the 14 items where all five transcriptions disagreed was wrong** – precision 1.00 as an abstention rule, recall 0.24 of all 58 errors. On $n = 100$ a zero-false-positive cell of size 14 is suggestive rather than established, but it is the most directly actionable thing in the pilot.

4 Reasoning Entropy: Clean Pipeline, Unresolved Signal

Unlike perception, the reasoning/grading pipeline required no comparable fix. It clusters a single parsed binary digit (error present/absent) rather than free text, so it never had the "wrong span" problem. The pipeline was independently re-verified by recomputing correctness from scratch for all 100 items against the stored results: zero mismatches.

4.1 $K = 5$ baseline

Metric	Value
AUROC (reasoning entropy $\rightarrow$ grading error)	0.618
95% CI	$[0.489, 0.741]$
Grading accuracy	75%
Median entropy, correct group ($n = 75$)	0.5004
Median entropy, incorrect group ($n = 25$)	0.5004

This interval includes 0.5. With 25 items in the minority class, an AUROC of 0.618 is not distinguishable from chance, and earlier drafts of this report describing it as a "real but weak" signal overstated it. Everything downstream in this section was motivated by trying to strengthen a number that was never established in the first place.

The medians are also *exactly* tied. With only 5 samples of a binary outcome, there are just 3 realistic entropy values (unanimous, a 4–1 split, a 3–2 split), which structurally limits how well two groups can be told apart by this statistic.

4.2 $K = 15$ follow-up: is the weak signal a real ceiling, or under-sampling?

To test whether $K = 5$ was simply too coarse to reveal a real effect, grading was resampled to $K = 15$ for the same 100 items (reusing the original 5 samples and drawing 10 more), keeping perception untouched.

Metric	$K = 5$	$K = 15$
AUROC	0.618	0.665
95% CI	$[0.489, 0.741]$	$[0.514, 0.800]$
Grading accuracy	75%	82%
Distinct entropy values reached	5	13
Median entropy, correct group	0.500	0.439
Median entropy, incorrect group	0.500	0.608

The exact median tie *does* break at $K = 15$ (Figure 2), and the $K = 15$ interval does clear 0.5 – though only just. A pre-registered decision rule (AUROC ≥ 0.72 *and* the median tie breaking, to count as "sharper"; 0.55–0.72 as "real but modest"; below 0.55 as "noise") classifies this as **confirmed_modest**.

That classification should be read with the paired comparison in mind, because the apparent $0.618 \rightarrow 0.665$ gain is *not* evidence that more sampling helped: the paired difference is -0.047 with a 95% CI of $[-0.178, +0.090]$, comfortably spanning zero. The tie breaking is a genuine structural consequence of having 13 reachable entropy values instead of 5, but tripling K bought no measurable discrimination. Note also that grading accuracy itself moved ($75\% \rightarrow 82\%$), so the two AUROCs are scored against different correctness labels – 25 errors vs. 18 – which the paired interval cannot correct for.

This pointed at a different question: is more sampling the right lever at all, or is the *digit itself* too coarse a readout of the model's judgment?

4.3 Reasoning-text clustering: a promising readout, an unproven improvement (unresolved)

The grading prompt asks the model to write a `**Reasoning:**` explanation before its `**Error:**` digit. Every analysis so far discarded that explanation and kept only the digit. Clustering the reasoning text instead – via the same bidirectional NLI entailment approach that failed for perception (Attempt 2) – was tested against the already-collected $K = 5$ raw samples, at no additional GPU cost.

Metric	Digit-only	Reasoning-text
AUROC	0.618	**0.702**
95% CI	$[0.489, 0.741]$	$[0.579, 0.817]$
Median entropy, correct group ($n = 75$)	0.500	0.950
Median entropy, incorrect group ($n = 25$)	0.500 (tied)	1.332
Paired difference (text − digit, same items, same labels): $+0.083$, 95% CI $[-0.064, +0.233]$		

The improvement is not statistically established. Reasoning-text clustering at $K = 5$ does clear chance on its own terms ($[0.579, 0.817]$ excludes 0.5) where the digit-only baseline does not ($[0.489, 0.741]$ does not) – but that pattern is precisely the "difference in significance is not a significant difference" trap. The correct test is the paired one, and its interval spans zero. An earlier draft of this report called this "a stronger, less noisy signal" and "arguably the most interesting result of the reasoning-arm investigation"; on the evidence available at $n = 100$, that claim is not supported. What survives is narrower and still worth having: reasoning-text clustering is the only reasoning-arm readout whose own interval excludes chance, and it has a documented mechanism (below) for why it should be less noisy than the digit. That makes it the right candidate to test at larger n, not a demonstrated win.

Given that the same NLI approach caused Attempt 2's over-merging failure on perception, this result was not trusted without directly checking for the same bias. The check: do any two samples that *disagree on the parsed digit* (one says error, one says no error) ever get merged into the same reasoning-text cluster? They do – 55 times across the 61 items where samples disagreed on the digit. Inspecting a diverse sample of these merges (8 cases, different items) showed 7 of 8 were not opposed reasoning merged by mistake: *both* texts explicitly argued "there is an error," just citing different specific details, while their own digits disagreed. Quantified across the full dataset: **60/494 samples (12%)** have digit$= 0$ while their own reasoning opens by flagging a mistake ("incorrect," "error," "mistake," ...). The model is self-contradicting within single samples – writing a reasoning that argues for an error, then emitting the opposite digit anyway, plausibly decoding noise on the final token, independent of the fully-formed judgment already written. At $K = 5$, reasoning-text clustering is not an artifact of NLI bias; it is a readout that is not vulnerable to this specific noise source the way the digit alone is (Figure 3).

This 12% self-contradiction rate is a measurement about the model, not about the clustering method, and it is unaffected by everything the confidence intervals overturn – it is a direct count over 494 parsed samples, not an AUROC comparison. It stands as an independent finding regardless of whether reasoning-text entropy ever proves to be the better predictor.

4.4 $K = 15$ GPU confirmation: the $K = 5$ result does not generalize (failed)

A GPU-accelerated confirmation at $K = 15$ (impractical on local CPU – a single item's 105 pairwise NLI comparisons took 35s locally; GPU reduced the full 100-item pass to 58s) was run via a dedicated notebook (`pilot/02_reasoning_text_entropy.ipynb`). It reproduced the $K = 5$ result almost exactly (AUROC 0.7019 vs. the 0.702 found locally), then revealed the $K = 15$ result does not hold up:

Metric	$K = 5$	$K = 15$
AUROC (reasoning-text)	0.702	**0.520**
95% CI	$[0.579, 0.817]$	$[0.378, 0.664]$
Items with a digit split	61/100	87/100
Cross-digit cluster merges	55	**1187**
Paired difference ($K = 5 - K = 15$, same items): $+0.182$, 95% CI $[+0.037, +0.330]$		

The $K = 15$ interval includes 0.5, so the technique is not distinguishable from chance at that K. More usefully, the degradation from $K = 5$ to $K = 15$ is the *only* reasoning-arm comparison in this report whose paired interval excludes zero – this is the one place where the data can actually tell two conditions apart. It carries one caveat that the interval cannot absorb: grading accuracy differs between the two runs (25 errors vs. 18), so the two AUROCs are scored against different label sets. The mechanistic evidence below is what makes the degradation credible rather than the interval alone.

The paired comparison against the digit-only baseline at $K = 15$ is -0.145, 95% CI $[-0.317, +0.037]$: directionally worse, not resolvably so. The cross-digit merge count grew far faster than the $\sim 10\times$ increase in available pairs ($\binom{15}{2} = 105$ vs. $\binom{5}{2} = 10$) would predict – a $\sim 21\times$ increase in merges from a $\sim 10\times$ increase in comparisons. Re-clustering individual items locally to inspect actual cluster membership found the cause: **transitive cascade merging**. The clustering method unions any two samples whose reasoning mutually entails, then merges transitively (union-find) – so a single mistaken pairwise link can silently absorb an entire subgroup into a cluster that does not actually agree. One item's largest cluster contained 11 of 15 samples, mixing digit=1 samples arguing *"incorrectly multiplies by 5 instead of 9/5"* with digit=0 samples arguing *"does not contain any apparent...errors"* – genuinely opposed conclusions, merged together. This is not universal: a second inspected item's largest cluster (7/15 samples) was entirely digit=0 with consistent reasoning, correctly merged. The failure is concentrated in a subset of items, but at $K = 15$ that subset is large enough to collapse the aggregate signal to near-random. More pairwise comparisons mean more chances for a single false-positive entailment call to cascade, and union-find has no mechanism to catch or bound that once it happens.

The mechanism is established: the cascade was found by re-clustering real items and reading actual cluster membership, and it reproduces as a unit test. What the confidence intervals add is a bound on how much of the *aggregate* $0.702 \to 0.520$ drop can be attributed to it – the paired interval $[+0.037, +0.330]$ is wide, so the degradation is real in direction but poorly determined in magnitude. A structural fix (e.g. requiring a minimum fraction of pairwise agreement within a cluster rather than any transitive chain, or a stricter/larger NLI model) would need to precede any further use of this technique at higher K – though at $n = 100$ such a fix could not be evaluated with any precision either, which is why sample size is the recommended next lever rather than the algorithm.

5 Integrity Checks

Both entropy signals depend on the sampling and parsing pipeline actually working. Two checks validate that before trusting any entropy number.

Check	Result
Temperature-0 anchor, transcription (2 greedy draws/item, must be ≈ 0)	0/100 non-zero
Temperature-0 anchor, grading (2 greedy draws/item, must be ≈ 0)	0/100 non-zero
Transcription parse-failure rate (before marker-regex fix)	13.4% (67/500)
Transcription parse-failure rate (after marker-regex fix)	4.0% (20/500)
Grading parse-failure rate ($K = 5$)	1.0% (5/500)
Grading parse-failure rate ($K = 15$)	0.9% (14/1500)

The temperature-0 anchor passing at 0/100 confirms the sampling/parsing pipeline is behaving deterministically at temperature 0, as it must. The transcription parse-failure rate was itself a real bug (the extraction regex only matched the exact `**Answer:**` markdown the prompt requested, missing common real-world drift such as LaTeX `\textbf{}` bold, plain text, and a full-width colon); widening the pattern recovered most of the missed samples.

6 Summary

Arm	Status	AUROC (95% CI)	Notes
Perception (final)	**Working**	**0.788** [0.697, 0.869]	Robust to all artifact cuts
Perception (attempt 1)	**Failed**	–	83% pinned at max entropy
Perception (attempt 2)	**Failed**	–	Over-merged, 95% implausible
Reasoning, digit ($K = 5$)	**Unresolved**	0.618 [0.489, 0.741]	CI includes chance
Reasoning, digit ($K = 15$)	**Unresolved**	0.665 [0.514, 0.800]	No paired gain over $K = 5$
Reasoning, text ($K = 5$)	**Unresolved**	0.702 [0.579, 0.817]	Beats chance, not the baseline
Reasoning, text ($K = 15$)	**Failed**	0.520 [0.378, 0.664]	Cascade over-merging

Reading the table by point estimate alone reverses two of its conclusions, so the paired comparisons are collected here explicitly. Exactly one of the five differences this pilot examined can be resolved at $n = 100$ – and it is a degradation, not a gain:

Comparison	Paired difference (95% CI)	Resolved?
Reasoning text − digit, $K = 5$	+0.083 [−0.064, +0.233]	No
Reasoning text − digit, $K = 15$	−0.145 [−0.317, +0.037]	No
Reasoning digit, $K = 5 - K = 15$	−0.047 [−0.178, +0.090]	No
Reasoning text, $K = 5 - K = 15$	+0.182 [+0.037, +0.330]	**Yes**
Perception − best reasoning arm	+0.086 [−0.073, +0.243]	No[†]

† Perception and reasoning predict errors on *different tasks* with different base rates, so this row is descriptive only – a paired interval is not the right instrument for it. Perception's standing rests on its own CI excluding chance and surviving §3.4's cuts, not on beating the reasoning arm.

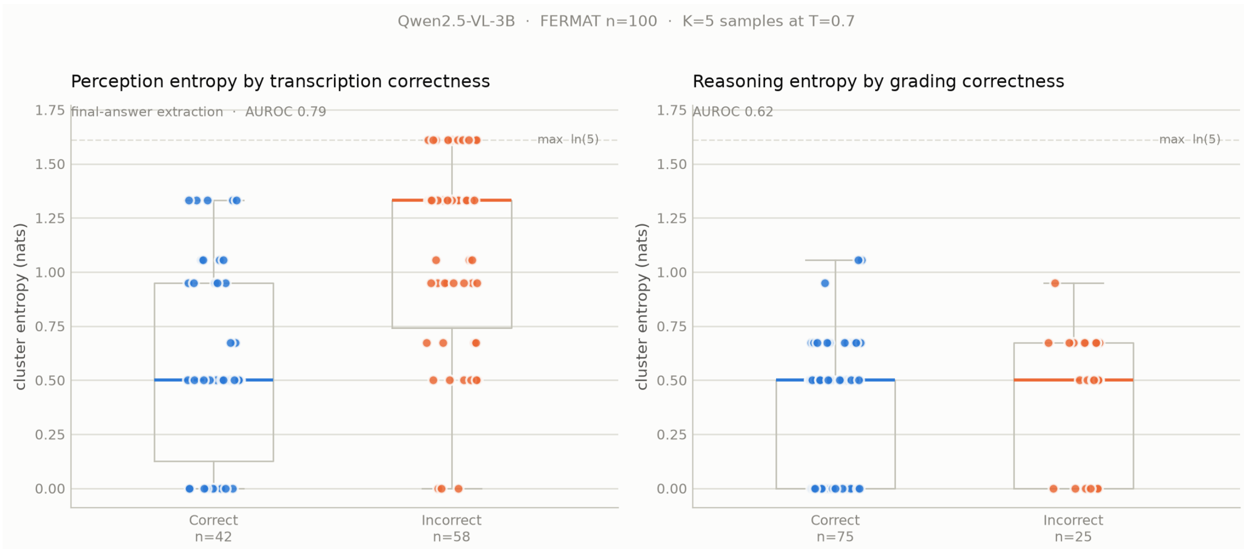

Figure 1: Final result for both arms, $n = 100$. Left: perception entropy by transcription correctness (AUROC 0.788, 95% CI $[0.697, 0.869]$). Right: reasoning entropy by grading correctness at $K = 5$ (AUROC 0.618, 95% CI $[0.489, 0.741]$ – includes chance).

7 Limitations

- $n = 100$ **is the binding constraint on every open question.** This is the central limitation, and it subsumes several others. With a minority class of 18–58 items, the 95% CI on a single AUROC spans roughly ± 0.12 and on a paired difference roughly ± 0.15. Four of the five comparisons this pilot ran are therefore unresolvable, including the two it was designed to answer. Any further work that does not increase n – a larger model, a fixed clustering algorithm, a new prompt design – would produce another point estimate with the same error bar and would not settle anything.
- **Single model.** All results are Qwen2.5-VL-3B; no comparison to a larger model (7B) or to a baseline uncertainty method.
- **Confidence intervals were added retrospectively.** The experiments were designed, run, and initially written up without them; the intervals were computed afterwards on the same data. They correct the report's claims but cannot retroactively supply the statistical power the experiments were never sized for.
- **Perception correctness label is strict.** Transcription correctness requires the extracted final answer to match ground truth after canonicalization; some "incorrect" classifications are plausibly extraction-span mismatches (the model's answer and the ground truth landing in different canonicalization tiers) rather than genuine disagreement, which would mean the true accuracy is somewhat higher than 42%. This biases the reported accuracy conservatively, not favorably.

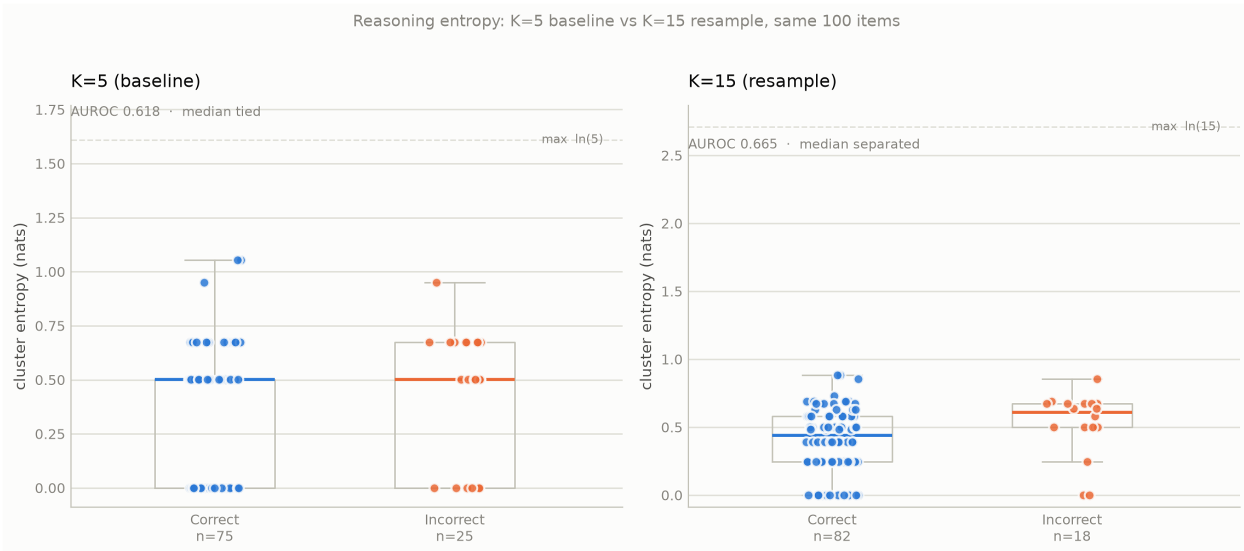

Figure 2: Reasoning entropy before ($K = 5$) and after ($K = 15$) resampling, same 100 items. The exact median tie at $K = 5$ breaks at $K = 15$ – a structural consequence of 13 reachable entropy values instead of 5 – but the paired AUROC difference is -0.047, 95% CI $[-0.178, +0.090]$: no measurable gain in discrimination.

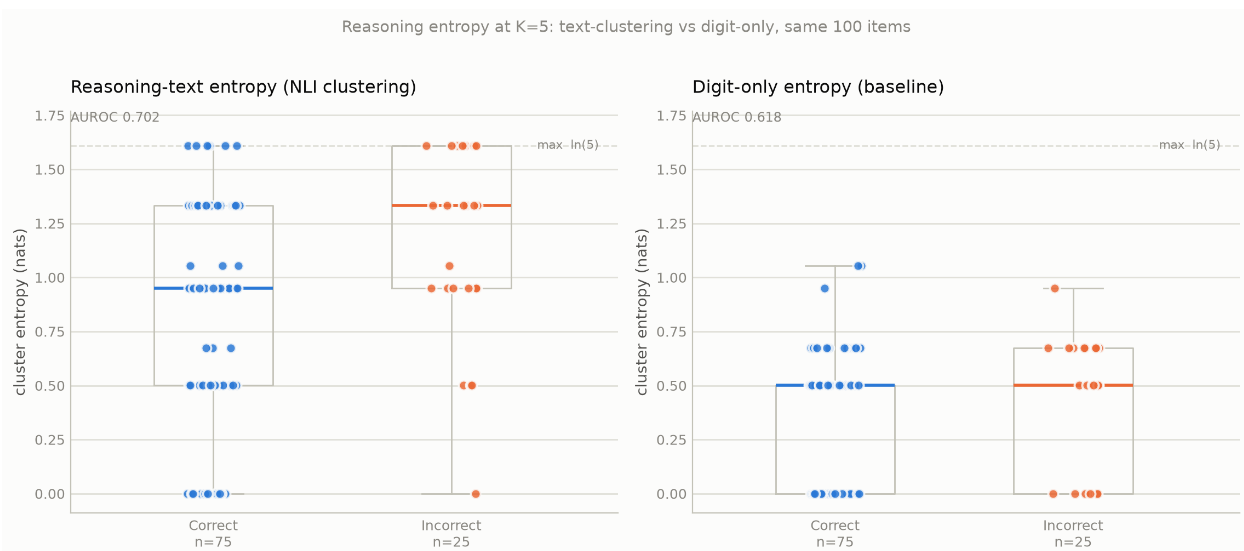

Figure 3: Reasoning entropy at $K = 5$: clustering the model's reasoning text (left, AUROC 0.702) vs. the digit-only baseline (right, AUROC 0.618), same 100 items. The reasoning-text approach separates the incorrect group's median above the correct group's; the digit-only approach ties them exactly. The visible gap is not statistically resolved, however: the paired difference is $+0.083$, 95% CI $[-0.064, +0.233]$ (§4.3).

- **No human double-grading.** Ground-truth labels are taken as given from the dataset.
- **Reasoning entropy's ceiling is model- and task-specific.** The binary-outcome information limit applies to *this* grading task design; a differently-posed prompt (e.g. eliciting a confidence score rather than a forced binary decision) was not tested.
- **Reasoning-text clustering has a confirmed mechanical defect at $K = 15$.** Tested and found to collapse to near-random (AUROC 0.520) due to transitive cascade over-merging (§4.4). The mechanism is verified by direct inspection of cluster membership; the magnitude of its effect on aggregate AUROC is not well determined at this n. The underlying cross-encoder is also a small, general-purpose NLI model rather than one tuned for mathematical reasoning text, which may independently limit how far this approach can go even after the cascade issue is fixed.

8 Recommended Next Steps

The ordering below follows from the limitation above: with four of five comparisons unresolvable at $n = 100$, increasing n dominates every other available lever, and should precede them rather than follow them.

1. **Scale the existing 3B pipeline to 300–500 items.** This is the highest-value next run and the cheapest of the scaling options: the pipeline already works, no quantization or model-loading changes are needed, and the cost is roughly linear in items. At $n = 400$ the CI on a single AUROC narrows from about ± 0.12 to about ± 0.06, which is the difference between "suggestive" and "settled" for every reasoning comparison in this report. Re-running the same five comparisons at that n requires no new analysis code – `pilot.plotting` already implements all of it.
2. **Dump the pairwise NLI entailment matrix on the next GPU run.** A cheap byproduct with disproportionate value: the expensive part of any clustering variant is the $O(K^2)$ NLI inference, so persisting that matrix once makes every future merge-criterion experiment a free, instant, offline computation. Do this regardless of which other step is taken.
3. **Then fix the clustering cascade fragility.** Replace or constrain the union-find transitive merge – e.g. require a minimum fraction of pairwise agreement within a cluster (a clique-style criterion) rather than any single chain of pairwise links, so one false-positive entailment call cannot silently absorb a whole subgroup. Deliberately sequenced after the scale-up: the fix is principled and cheap, but its effect could not be measured against a ± 0.15 error bar, and evaluating several candidate criteria against $n = 100$ would amount to fitting the merge rule to noise.
4. **Token-level confidence as an alternative reasoning signal.** Instead of inferring uncertainty from repeated discrete sampling, use the model's own next-token probability for the 0/1 decision at generation time. A richer, more direct signal, immune to the clustering-cascade problem entirely since it does not require sample-to-sample comparison – but requires new sampling code and a new run to validate.
5. **Compare against a stronger model (7B) last.** A 7B run at $n = 100$ would return a second point estimate with the same ± 0.12 interval as the first, which could not be distinguished from the 3B result – the comparison only becomes informative once n supports it.